VOLUME I: GENERAL OPERATING CONDITIONS & BASELINE PIPING SCHEDULES

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1.0 GENERAL SITE PARAMETERS & PRECEDENCE HIERARCHY

1.1 Facility Classification and Exposure

The Borealis Central Processing Facility and all associated satellite gathering manifolds (Specifically Manifold Loops A, B, and C) are designated as a **Class-3 Arctic Industrial Zone (Zone-A3)**. Unlike enclosed temperate skid modules, all primary manifold assemblies, pigging receivers, and outdoor transfer headers operate continuously in exposed ambient conditions subject to wind-chill and radiative cooling.

1.2 Thermal & Pressure Envelope Baseline

Under normal steady-state circulation, the primary process fluid is maintained between **+15°C and +40°C** at a nominal operating pressure of **120 PSI (8.27 bar)**, with surge tolerances up to 145 PSI. However, meteorological records for Zone-A3 mandate a winter minimum ambient design temperature (MDMT) of **-42°C (-43.6°F)** during shutdown, stagnant hold, or cold-restart events. Consequently, any piping, valve, fastener, or instrumentation component installed in Zone-A3 is strictly subject to the *Cryogenic Excursion Protocol* detailed in Volume II, Section 4.3.

1.3 Order of Precedence for Conflicting Specifications

Contractors and automated quoting systems must resolve material conflicts using a strict three-tier precedence model. In the event of conflicting part numbers, material grades, or pressure-temperature ratings:

- **Tier 1 (Baseline):** Section 2.0 Baseline Piping & Valve Schedules apply only to standard non-excursion service.
- **Tier 2 (Environmental & Chemical Overrides):** Volume II, Section 4.0 Environmental and Chemical Excursion Protocols strictly **SUPERSEDE** all Tier 1 Baseline Schedules whenever their triggering thresholds are met.
- **Tier 3 (Active Safety Interlocks):** Volume II, Section 5.0 Mechanical & Actuator Safety Interlocks **SUPERSEDE** both Tier 1 and Tier 2 specifications.

2.0 BASELINE PIPING, VALVES & FASTENER SCHEDULES

The following baseline schedules govern standard 100 PSI to 150 PSI utility and process transfer lines prior to applying site-specific environmental or chemical overrides.

2.1 Baseline Mainline & Header Piping

For pressurized fluid transfer headers operating between 100 PSI and 150 PSI, standard High-Density Polyethylene (HDPE) **PE-4710 SDR-11** piping (Part #**PIPE-HDPE11-STD**) with butt-fusion joints is approved as the default baseline material.

2.2 Baseline Manifold Isolation Valves & Elastomer Seals

Standard manifold isolation and block valves across Loops A, B, and C shall utilize 316 Stainless Steel floating ball valves, Part #**VLV-200-SS**, factory-fitted with standard fluorocarbon Viton (FKM-A) O-ring stem and seat seals (Part #**SEAL-VT-09**), paired with standard pneumatic rack-and-pinion actuator Part #**ACT-PN-100**.

Service Loop	Nominal Pressure	Baseline Pipe SKU	Baseline Valve SKU	Baseline Seal SKU	Baseline Actuator SKU
Manifold Loop A (Glycol Return)	110 PSI	PIPE-HDPE11-STD	VLV-200-SS	SEAL-VT-09 (FKM-A)	ACT-PN-100
Manifold Loop B (Primary Transfer)	120 PSI	PIPE-HDPE11-STD	VLV-200-SS	SEAL-VT-09 (FKM-A)	ACT-PN-100
Manifold Loop C (Produced Water)	95 PSI	PIPE-HDPE11-STD	VLV-200-SS	SEAL-VT-09 (FKM-A)	ACT-PN-100

**Table 2.1 Note: All baseline SKUs listed above assume +15°C minimum ambient exposure and zero polar-solvent dosing. Verify chemical dosing streams in Section 3.0 and excursion overrides in Volume II before releasing Bill of Materials.*

2.3 Baseline Flange Fasteners & Bolting

All flanged metallic-to-polymer transition spools and valve bodies shall be assembled using **ASTM A193 Grade B7** chromium-molybdenum alloy steel stud bolts paired with **ASTM A194 Grade 2H** heavy hex nuts (Kit #**BOLT-B7-2H**), unless overridden by low-temperature brittle-fracture schedules (see Section 4.2).

3.0 PROCESS FLUID COMPOSITION & INJECTION STREAMS

3.1 Primary Heat-Transfer Stream

The primary heat-transfer medium circulating through Manifold Loops A and B consists of a 60/40 inhibited propylene glycol/water mixture containing up to 1.5% entrained light hydrocarbon condensate.

3.2 Automated Anti-Hydrate Methanol Injection (Manifold Loop B)

To prevent gas-hydrate plugging during winter operations, whenever ambient temperatures fall below **-15°C**, Automated Chemical Dosing Skid #DS-04 continuously injects **anhydrous methanol (MeOH)** at 8.0% volume concentration directly into **Manifold Loop B** upstream of the primary isolation valves. Manifold Loops A and C do not receive methanol injection.

3.3 Polar-Solvent Service Mandate: Any piping segment, valve chamber, or elastomer seal exposed to anhydrous methanol injection (as defined in Section 3.2) is formally classified as a **Polar-Solvent Service Line**. All components in a Polar-Solvent Service Line must comply with the chemical compatibility overrides in Volume II, Section 4.5.